

Model Question Paper-I with effect from 2025

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1BMATE101

First Semester B.E./B.Tech. Degree Examination Differential Calculus & Linear Algebra

TIME: 03Hours

Max.Marks:100

- Note: 1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
 2. VTU Formula Hand Book is Permitted
 3. M: Marks, L: Bloom's level, C: Course outcomes

Module-1			M	L	C
Q 1.	a	With usual notations, prove that $\tan\phi = r \frac{d\theta}{dr}$.	6	L2	1
	b	Show that the angle of intersection of the curves $r = a \log \theta$ and $r = a/\log \theta$ is $\tan^{-1} \left[\frac{2e}{(1-e^2)} \right]$.	7	L2	1
	c	Show that the radius of curvature at $x = \frac{\pi}{2}$ of the curve $y = 4\sin x - \sin 2x$ is $\frac{5\sqrt{5}}{4}$.	7	L2	1
OR					
Q 2.	a	With usual notations, prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$.	6	L2	1
	b	Find the angle of intersection of the curves $r^2 \sin 2\theta = 4$ and $r^2 = 16 \sin 2\theta$	7	L2	1
	c	Find the radius of curvature using pedal form for the curves, $r^2 \sec 2\theta = a^2$	7	L2	1
Module-2					
Q 3.	a	Expand $\log(\sec x)$ in powers of x as far as the term in x^4 .	6	L2	1
	b	If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ prove that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.	7	L2	1
	c	Examine the function for extreme values $f(x, y) = x^3 y^2 (1 - x - y)$.	7	L2	1
OR					
Q 4.	a	Evaluate $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan(\pi x/2a)}$	6	L2	1
	b	If $x + y + z = u$; $y + z = v$; $z = uvw$ then find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$	7	L2	1
	c	Show that a differential equation for the current i in an electrical circuit containing an inductance L and resistance R in series and acted on by an electromotive force $E \sin \omega t$, satisfies the equation $L \frac{di}{dt} + Ri = E \sin \omega t$. Find the value of the current at any time t, if initially there is no current in the circuit.	7	L3	1
Module-3					
Q 5.	a	Solve $(xy^2 - e^{1/x^3}) dx - x^2 y dy = 0$	6	L2	1
	b	Solve for p. $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$.	7	L2	1

	c	A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction.	7	L3	1
OR					
Q 6.	a	Solve $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$	6	L2	1
	b	Show that the family of parabolas $y^2 = 4a(x + a)$ is self-orthogonal.	7	L2	1
	c	Solve the equation $(px - y)(py + x) = 2p$ by reducing into Clairault's form, by taking the substitutions $u = x^2$ and $v = y^2$.	7	L2	1
Module-4					
Q 7.	a	Solve $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{-x} + \sin 2x$	6	L2	1
	b	Solve by variation of parameter method $\frac{d^2y}{dx^2} + y = \frac{1}{1 + \sin x}$	7	L2	1
	c	Solve $x^2y'' + xy' + y = 2\cos^2(\log x)$	7	L2	1
OR					
Q 8.	a	Solve $\frac{d^2y}{dx^2} - 4y = 3^x + \cosh(2x - 1)$	6	L2	1
	b	Solve $(3 + 2x)^2 \frac{d^2y}{dx^2} + 5(3 + 2x) \frac{dy}{dx} + y = 4x$	7	L2	1
	c	Solve by variation of parameter method $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = \frac{1}{1 - e^x}$	7	L2	1
Module-5					
Q 9.	a	Find the rank of the following matrices by reducing into row echelon form. $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$	6	L2	2
	b	Solve by Gauss – Seidel iteration method: $20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25$	7	L2	2
	c	Determine the dominant Eigen value and the corresponding Eigen vector of the following matrix by using Rayleigh's power method. $\begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$	7	L2	2
OR					
Q 10.	a	Investigate the values of λ and μ so that the equation $2x + 3y + 5z = 9; 7x + 3y - 2z = 8; 2x + 3y + \lambda z = \mu$ have (i) no solution (ii) a unique solution and (iii) an infinite number of solution.	6	L2	2
	b	Solve by Gauss elimination method: $2x + y + z = 10; 3x + 2y + 3z = 18; x + 4y + 9z = 16.$	7	L2	2
	c	Find the Eigen values and Eigen vectors of the following matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	7	L2	2

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1BMATE101

First Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Differential Calculus and Linear Algebra : EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. VTU Handbook is permitted.*

Module - 1			M	L	C
Q.1	a.	With usual notations, prove that $\tan \theta = r \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find angle between the following curves $r = a \log \theta$ & $r = \frac{a}{\log \theta}$.	7	L2	CO1
	c.	Show that the radius of curvature of the curve $r^n = a^n \cos n\theta$ varies inversely as r^{n-1} .	7	L2	CO1
OR					
Q.2	a.	Find the radius of curvature of the curve $x^3 + y^3 = 3axy$ at the point $(3a/2, 3a/2)$.	6	L2	CO1
	b.	Show that the following pair of curves cut orthogonally: $r = a(1 + \sin \theta)$ & $r = a(1 - \sin \theta)$	7	L2	CO1
	c.	Find the pedal equation $r^2 = a^2 \sec 2\theta$.	7	L2	CO1
Module - 2					
Q.3	a.	Expand $\log(\sec x)$ in powers of x up to the term containing x^4 .	6	L2	CO1
	b.	Evaluate i) $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{1/x}$ ii) $\lim_{x \rightarrow \pi/2} (\sec x)^{\cot x}$.	7	L2	CO1
	c.	Discuss the maxima and minima of the function $f(x, y) = x^3 + y^3 - 3x - 27y + 24$.	7	L3	CO1
OR					
Q.4	a.	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.	6	L2	CO1
	b.	If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$ evaluate Jacobian of u, v, w with respect to x, y, z .	7	L3	CO1
	c.	If $u = x^2 + y^2 + z^2$, $x = e^{2t}$, $y = \cos 3t$, $z = \sin 3t$. Find $\frac{du}{dt}$.	7	L2	CO1
Module - 3					
Q.5	a.	Solve $(y \cos x + \sin y + y)dx + (\sin x + x \cos y + x)dy = 0$.	6	L2	CO1
	b.	Find a solution for the non-linear differential equation $xyp^2 - (x^2 + y^2)p + xy = 0$.	7	L2	CO1
	c.	Solve $(px - y)(py + x) = 2p$ by reducing into Clairaut's form, taking the substitutions $x^2 = X$, $y^2 = Y$.	7	L2	CO1
OR					
Q.6	a.	Solve $\frac{dy}{dx} + y \tan x = y^3 \sec x$.	6	L2	CO1

	b.	Solve $(xy + y^2 + y)dx + (x^2 + 3xy + 2x)dy = 0$.	7	L2	CO1
	c.	Find the orthogonal trajectories of curve $r^n = a^n \sin n\theta$.	7	L3	CO1
Module - 4					
Q.7	a.	Solve $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 4y = 0$.	6	L2	CO1
	b.	Solve $x^2 \frac{d^2y}{dx^2} - 2y = x^2 + \frac{1}{x}$.	7	L2	CO1
	c.	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \tan x$.	7	L2	CO1
OR					
Q.8	a.	Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 12x^2$.	6	L2	CO1
	b.	Solve $y'' - 6y' + 9y = 5e^{3x} - \sin 2x$.	7	L2	CO1
	c.	Solve $(3x + 2)^2 y'' + 3(3x + 2)y' - 36y = 8x$.	7	L2	CO1
Module - 5					
Q.9	a.	Determine the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$.	6	L2	CO2
	b.	Solve the following system of equations by Gauss-Seidal $10x + y + z = 12$, $x + 10y + z = 12$, $x + y + 10z = 12$, Carry out three iteration.	7	L2	CO2
	c.	Find the dominant eigen value and the corresponding eigen vector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ by power method. With initial Eigen vector $[1 \ 0 \ 0]^T$. Carry out four iterations.	7	L3	CO2
OR					
Q.10	a.	Solve by Gauss elimination method $x + y + z = 9$, $x - 2y + 3z = 8$, $2x + y - z = 3$.	6	L2	CO2
	b.	Investigate the values of λ and μ , such that the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have i) Unique solution ii) No solution iii) Infinitely many solution.	7	L3	CO2
	c.	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.	7	L2	CO2

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First Semester B.E/B.Tech. Degree Examination, June/July 2026
Differential Calculus and Linear Algebra: EEE Stream

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU formula Hand Book is permitted.

Module - 1				M	L	C
1	a.	With usual notation prove that $\tan \phi r = \frac{d\theta}{dr}$.		6	L2	CO1
	b.	Find angle between the following curves $r = \sin\theta + \cos\theta$ and $r = 2\sin\theta$.		7	L2	CO1
	c.	Find the radius of curvature for the curve at $(2a, 0)$ on the curve $y^2 = \frac{4a^2(2a-x)}{x}$.		7	L2	CO1
OR						
2	a.	With usual notation prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$		6	L2	CO1
	b.	Show that the following pairs of curves cut orthogonally $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$		7	L2	CO1
	c.	Show that for the curve $r(1 - \cos\theta) = 2a$, p^2 varies as r^3 .		7	L2	CO1
Module - 2						
3	a.	Expand $\sqrt{1 + \sin 2x}$ in powers of x as far as the term in x^4 .		6	L2	CO1
	b.	If $u = f(y - z, z - x, x - y)$, then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.		7	L2	CO1
	c.	Examine the extreme values of the following function: $f(x, y) = x^3 + y^3 - 3x - 12y + 20$		7	L2	CO1
OR						
4	a.	Evaluate: i) $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}}$ ii) $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$		6	L2	CO1

	b.	If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$, then show that $J = \begin{pmatrix} u, v, w \\ x, y, z \end{pmatrix} = 4$	7	L2	CO1
	c.	Find the total derivative of u as a function of t for $u = xy + yz + zx$ where $x = t \cos t$, $y = t \sin t$, $z = t$	7	L2	CO1
Module – 3					
5	a.	Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$.	6	L2	CO1
	b.	Solve $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$.	7	L2	CO1
	c.	Find the orthogonal trajectory for the family of curves $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$.	7	L2	CO1
OR					
6	a.	Solve $\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$.	6	L2	CO1
	b.	Solve $(px - y)(py + x) = 2p$ by reducing into Clairaut's form taking the substitution $x^2 = X$, $y^2 = Y$.	7	L2	CO1
	c.	Solve $(x^2 + y^2 + x)dx + xy dy = 0$.	7	L2	CO1
Module – 4					
7	a.	Solve $\frac{d^3 y}{dx^3} - 3 \frac{dy}{dx} + 2y = 0$.	6	L2	CO1
	b.	Solve $y'' + 2y' + y = 2x + x^2$.	7	L2	CO1
	c.	Solve $x^2 y'' + xy' + 9y = 3x^2 + \sin(3 \log x)$.	7	L2	CO1
OR					
8	a.	Solve by the method of variation of parameter $y'' + 4y = \tan 2x$	6	L2	CO1
	b.	Solve $\frac{d^3 y}{dx^3} + y = 65 \cos(2x + 1)$.	7	L2	CO1
	c.	Solve $(2x + 1)^2 y'' - 6(2x + 1) y' + 16y = 8(2x + 1)^2$	7	L2	CO1

Module – 5

9	a.	Determine the rank of the matrix	6	L2	CO2
		$\begin{bmatrix} 91 & 92 & 93 & 94 & 95 \\ 92 & 93 & 94 & 95 & 96 \\ 93 & 94 & 95 & 96 & 97 \\ 94 & 95 & 96 & 97 & 98 \\ 95 & 96 & 97 & 98 & 99 \end{bmatrix}$			
	b.	Solve the following system of equations by Gauss Seidal method $20x + y - 2z = 17$ $3x + 20y$ $-z = -18$ $2x - 3y + 20z = 25$	7	L2	CO2
	c.	Find the eigen value and eigen vectors of the following matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	7	L2	CO2
OR					
10	a.	Solve the following system of equations by using Gauss Elimination method $2x + y + 4z = 12$ $4x + 11y$ $-z = 33$ $8x - 3y + 2z = 20$	6	L2	CO2
	b.	Investigate the values of λ and μ such that the system of equations $x + y + z = 6$ $x + 2y + 3z = 10$ $x + 2y + \lambda z = \mu$	7	L2	CO2
	c.	Find the largest eigen value and the corresponding eigen vector of the matrix A by the power method given that $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$	7	L2	CO2
